Since the nation's inception in 1776, it has been plagues by at least 10 financial crisis and even though they may seem unrelated at first glance, a deep dive into their history, causes, and effects shed light on the fact that they have a plethora of similarities. When we're comparing the 1907 and 1929 crises with the 2008 GFC, there are many resemblances that I will highlight in this paper while also comparing their histories, causes, and effects. Firstly, all 3 of these crises were preceded by a period of rapid economic growth and an increased amount of money and cash equivalents flowing in the economy. In the early 1900s, novel gold mining techniques emerged, leading to an increase in worldwide gold production and resulting in additional funds. Prior to the 1929 crisis the British economy was struggling and in an effort to bolster sterling the US government lowered interest rates. This not only improved the value of the sterling but also made domestic borrowing cheaper. In relation to GFC, Alan Greenspan reduced interest rates to help revitalize the economy following the dot-com bubble burst and 9/11. Additionally, the Community Reinvestment Act made it easier for people with subprime credit to obtain loans and the Gramm-Leach-Bliley Act which encouraged banks to grant more subprime loans, and the created LIAR loans. Owing to their name liar loans allowed individuals to borrow money without providing proof of income which meant that a greater proportion of the population now had access to lending capabilities which increased money flowing in the economy. Furthermore, all of these financial crises were instigated by various financial advancements that offered individuals more investment prospects. In the 1907 crisis had the introduction of life insurance and unregulated trust companies that functioned similarly to banks but guaranteed higher interest rates. Conversely, the 1929 crisis witnessed the creation of leveraged financial trusts, which enabled individuals to borrow funds to purchase stock by employing their own stock as collateral, resulting in highly leveraged portfolios. Lastly, the 2008 crisis witnessed the emergence of multiple products related to the highly leveraged housing market including mortgage-backed securities (MBSs), collateralized debt obligations (CDOs), and LIAR loans. Similarly, the establishment of the Fannie & Freddie firm created a secondary market for investing in housing and made it simpler for people to speculate. Thus, a combination of easy credit and intense speculation led to the formation of the bubble and GFC subsequently.

Another significant similarity among the three crises is the presence of fraudulent individuals. During the 1907 crisis Augustus Heinze and Charles Morris attempted to corner the copper market resulting in a run-on

Knickerbocker bank and the subsequent disintegration of the banking system. In 1929 Joseph Kennedy, Albert Wiggins, and Charles Mitchell some of the people who manipulated the market leading to unforeseen euphoria in the markets. In the 2008 crisis individuals and institutions exploited the CRA and LIAR to acquire loans while rating agencies received bribes to issue high ratings to dangerous MBS and CDO type instruments. Bank runs and failures were also a common theme during all these crises and in 1907 Augustus Heinze's failed attempt to corner United Copper stocks resulted in the collapse of Mercantile Bank & the Knickerbocker Trust. The contagion rapidly spread to other banks, and soon the situation escalated into a full-blown financial crisis. Similarly, in 1929 the Bank of Tennessee failed after supporting a firm that speculated in the stock market and the Bank of United States collapsed after the market crash. This ultimately led to the failure of over 8000 banks in the United States across two years. In 2008 after Fannie and Freddie stopped issuing subprime loans, the primary subprime mortgage lender New Century declared bankruptcy and credit rating agencies downgraded Countrywide. Additionally, the failure of two hedge funds heavily invested in CDOs and backed by Bear Stearns triggered the decline of Bear Stearns' stock. Subsequently, banks such as Washington Mutual, Wachovia, Lehman, and Merrill started to fail and Goldman Sachs and Morgan Stanley were close to shutting down. The stock market had previously reached an all-time high but eventually crashed catastrophically in 2008 triggering GFC.

Lastly, all of these financial crises culminated in the introduction of new government regulations aimed at controlling the banking industry and preventing a similar crisis from ever happening again. The central bank was formed following the 1907 crisis and the Owen Glass Federal Reserve Act of 1913 established a federal note backed by credit and gold and was intended to safeguard the country against future financial crises. After the 1929 crisis, the Banking Act of 1932 was enacted and it insured deposits in cooperation with the FDIC, in order to prevent further bank runs. We also had the Securities Act that increased transparency in financial reporting and regulated IPOs but most importantly the Glass-Steagall Act of 1933 was introduced and it separated commercial banks from investment banks to protect the industry and prevent future crises. In 2008, the Fed took several actions to stabilize the financial system and non-bank financial companies were now subjected to more stringent regulations. Banks were now required to undergo annual stress tests, had greater capital requirements, and had to have a backup plan for dissolution. To save the economy and bring it out of a recession the Fed also lowered

interest rates to near zero rates and pumped around $750 billion into the system. Subsequently interest rates were kept at near zero rates for almost a decade and this caused a variety of problems to the economy that we are both directly and indirectly facing right now.

I also think that we are also in an asset bubble that is slowly popping because we are currently in its blow-off phase. The crypto market has melted down, home prices in major markets are dropping, investors in stocks and bonds are suffering and many analysts think there is more to come. Following the start of Covid that served as an exogenous shock, the immediate government intervention and the fiscal stimulus worsened pre-existing bubbles already building in the asset world. As famous Austrian economist said "The boom plants the seeds of its future destruction". Money supply (M2) drastically increased in March 2020 due to the massive Covid stimulus package and during the 2020-2021 period the Fed pumped trillions of dollars into the economy. On December 25th 2019, the securities portfolio of the Fed was $3.8 trillion and this has risen to $8.5 trillion by April 6th 2022. During this time, the Fed went through a period of quantitative easing in which it was purchasing $120 billion in securities every month. The Fed had to act at this time and most people agree its actions prevented the U.S economy from collapsing due. However, an asset bubble was created by extremely low interest rates and high levels of spending causing financial markets to boom to unprecedented levels. Between March 2020 and 2022, M2 surged by $5.8 trillion (41%) and during this same period the S&P increased by 47% while bitcoin skyrocketed by 410%. Thus, we can see that the stealth phase and awareness phase started early 2021 and the mania phase lasted from mid-2021 to 2022 as asset prices continually increased.

The Fed essentially fuelled a massive run up in financial asset prices by slashing interest rates to near zero rates and buying up trillions of dollars in bonds. Now, as the central bank has been aggressively approving rate hikes for the past year to curb inflation, the bubbles are blowing-off as asset prices return to pre-pandemic levels. Amid these rate hikes home prices dropped 2.6% this quarter and rising rates push bond prices down with one Bloomberg index falling ~7% year to date. Part of the reason that this rapid decline in asset prices has not led to worse consequences is the fact that these assets were dropping from such a high, almost artificial point and they are simply going back to normal levels now. According to Jamie Dimon when we were in the covid recession, the economy needed a cash influx and high asset prices to recover fast, but once it recovered the Fed "wanted to

take away that boom". Although this works in theory, in practice when you bring asset prices up and then take them down very quickly a bubble is formed and eventually bursts and this is exactly what is happening now. According to Charles Schwab American consumers have been shielded by the strength of the labour market: a stronger current labour market means more people are employed and they are less likely to sell their homes and assets to make ends meet. If unemployment was on the rise and large portions of the American public began selling their assets this would contribute to the already bursting bubble and asset prices would fall at a much faster rate. However, consumer savings have begun to show signs of reduction, the covid unemployment benefits are slowing down, and credit card debt has surged which suggest that there is a chance this happens in the near future.

US equity markets ended 2020 ~16% up and were ~27% up in 2021 due to the awareness and mania phase of the bubble when money was pumped into the economy. However, following rate hikes and quantitative tightening starting in March 2022, the S&P 500 shed 19.4%, Nasdaq was down 33.1%, and the Dow lost 9% in a year. As of 2023, the S&P is up around 5% and the NASDAQ has risen 10% but according to Richard Bernstein Advisors the bubble in stocks has burst and investors who are betting on a rally in the market are still in denial over the changing economy. Analysts claim that the "stock market rally so far this year seems based largely on speculation rather than fundamentals" and that investors appear hopeful that the Fed will soon return to a policy of cheap and abundant liquidity while ignoring the Fed's warnings to the same. While not burst yet, the real estate bubble is also in its blow off phase with a decline in housing prices. The inventory of homes on the market has increased from 850,000 in Feb 2022 to 1.4million in Feb 2023 according to the National Associate of Realtors (41% rise). Similarly, the aggressive rate hikes mean that many buyers are priced out of the housing market due to rising mortgage rates which further decreases demand and increases supply. With fewer qualified buyers and more homes on the market, prices have been steadily declining as the bubble is slowly bursting.

Similar to the 2008 crisis we also had a few banks fail including SVB and Signature Bank. CS was on the verge of collapsing till it was agreed to be acquired by UBS. SVB failed because it had drawn so much money into the bank and had no demand for loans. Similarly, it did not have a Chief Risk Officer and thus invested its excess money in longer term securities to increase its interest rate margin. When depositors began to withdraw money out of SVB over the past 6 months or so, SVB could only repay them by selling its longer term assets but

those assets had fallen in value due to aggressive rate hikes and bursting bubbles. As depositors began withdrawing more, SVB began selling more assets and eventually it failed.

In order to prevent or reduce the adverse impact of this bubble we need to address the banking failures it indirectly caused as well as the falling asset prices due to rate hikes. If the Fed can tame inflation without crashing the economy and achieve the 'soft landing' it would defy the beliefs of most economists and also reduce damage to broader financial markets. Although the recent unemployment and consumer spending data has been strong, inflation has already been brought to ~6%. Although this is far from the Fed's goal of 2% it is important to note the lag associated with monetary policy and we need a while for the economy to adjust. Similarly, it is more important to prevent a banking crisis and a further downward spiral in asset prices so I think the Fed should tame down its rate hikes for the rest of this year. Thus, the Fed should stick to either a 25 basis rate hike every few months instead of choosing to increase rates by 25-50bps every meeting. Additionally, in order to prevent fraudulent claiming of benefits that happened during Covid, the government should pass certain bills that does throughout background checks on citizens before they can claim their benefits. Similarly, excessive spending of stimulus and unemployment benefits that occurred during that era contributed to forming the asset bubble and the government should control what people can spend their benefit packages on and also reduce the amount of benefits they give directly to people. Targeted policies like improving education and employment opportunities would be a better way to spend government money without contributing to a bubble. Coming to the banking crisis, collateralizing SVB's toxic assets would prevent the crisis from spreading to other financial institutions. Two Nobel prize winners Doug Diamond and Philip Dybvig shed light on the fact that the Dodd-Frank rule leaves the financial system vulnerable and needs to be revamped. When it came to SVB, uninsured depositors were assured that they could get money out at the promised value even though many of the assets in which the bank invested would decline in value if interest rates rose. For small levels of withdrawals, SVB could offer redemptions by selling assets that held value, but at some point SVB would need to start selling impaired assets and this is what happened. Revamping old laws and forming and strengthening new bodies like the FSOC and SEC are the future.